

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 8 OF 9

Document 8 - Contextual, Geographic, Seasonal, and Street-Scene Realism

Location Authenticity, Weather, Architecture, Camera Height, and Background
Integration

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

Version 2.0 - April 2026

Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Teach AuthenticLens to analyze and create realistic environments, especially background replacements and street scenes. This document focuses on the ordinary details that make a scene believable: geography, season, architecture, traffic logic, sidewalks, weather, scale, and camera placement.

Intended Gemini Behaviour

- Verify requested locations when current accuracy matters.
- Do not invent famous landmarks or storefronts in impossible positions.
- Match subject camera angle, light, and lens to the replacement environment.
- Use season and weather cues consistently.
- Use plausible everyday street details instead of cinematic fantasy.

A. Contextual Realism Stack

Layer	Question to answer	Common failure
Geography	Does the scene resemble the requested city/neighbourhood?	Generic North American street with wrong landmarks.
Season	Do trees, clothing, pavement, snow, sky, and light match the date?	Winter scarves/snow in a spring scene without reason.
Weather	Are sky, pavement, reflections, clothing, and shadows consistent?	Wet pavement with dry shoes/no reflections.
Architecture	Do buildings, window heights, storefront spacing, and materials make sense?	Melted facades, fake signs, impossible door sizes.
Street layout	Are roads, sidewalks, curbs, bike lanes, crosswalks, vehicles, and pedestrians plausible?	Sidewalk at same level as road, floating curb, wrong vanishing points.
Camera integration	Does subject scale match camera height and lens?	Subject pasted at wrong horizon/scale.

B. Background Replacement Checklist

1. Extract original camera height from the subject: eye-level, chest-level, waist-level, low-angle, or high-angle.
2. Identify original lens behaviour: wide-angle distortion, normal perspective, telephoto compression, or phone deep focus.
3. Choose a background with the same horizon line and vanishing point logic.
4. Match light direction, colour temperature, and shadow softness.
5. Create contact shadows under shoes, wheels, bags, and nearby objects.
6. Add occlusion where subject intersects ground or foreground objects.
7. Match grain/noise/compression between subject and background.

8. Validate location details: signs, storefronts, road layout, lane markings, trees, weather, and architecture.
9. Remove impossible or distracting landmarks unless explicitly requested and geographically plausible.
10. Run final check for pasted edges, scale errors, and synthetic background textures.

C. Ottawa / Rideau Street Early Spring Example

Operational guidance: when the user requests Rideau Street, Ottawa, Canada, early spring / April 23, 2026, the model should produce a plausible urban Ottawa street environment rather than a tourist-poster collage. If exact storefronts or current construction status matter, research current references before generating.

Element	Recommended cue	Avoid
Season	Early spring: bare or budding trees, cool daylight, possible damp pavement, lighter jackets.	Deep snowbanks, heavy scarves, summer leaves, tropical lighting.
Architecture	Mid-rise downtown storefronts, transit/urban details, realistic window spacing.	Random European streets, impossible Parliament placement, fantasy skyscrapers.
Road/sidewalk	Clear curb separation, realistic sidewalk width, road perspective, crosswalk/lane logic.	Sidewalk and road at same height; floating curbs; road perspective not matching subject.
Light	Overcast or cool spring daylight unless specified; shadows soft if cloudy.	Cinematic sunset unless requested; studio light on subject with street background.
Signage	Use readable generic signs or blur distant signs. Do not generate garbled brand text.	Fake store names, pseudo-text, copyrighted logos that are unreadable.

D. Perspective and Scale Rules

- Feet define ground contact. If feet are visible, their soles must align with the same ground plane as nearby curbs and pedestrians.
- Head height defines relative scale. A subject cannot be the same height as a distant pedestrian unless depth explains it.
- Verticals should remain vertical unless lens tilt is intentionally visible.
- A wide-angle background exaggerates depth; a telephoto subject looks compressed. Do not mix them without matching blur and scale.
- Camera height controls horizon. Eye-level portraits place horizon near eye/upper torso zones; low-angle shots shift horizon lower.

E. Weather and Light Consistency

Weather/light condition	Required visual consistency
Overcast	Soft shadows, lower contrast, cooler sky, no hard

	cast shadows.
Sunny	Clear shadow direction, bright highlights, high contrast, stronger squint/catchlights.
Rain/wet pavement	Specular reflections on ground, darker pavement, possible umbrella/jacket cues, damp shoe contact.
Snow/winter	Snow on ground/curbs/roofs, colder clothing, diffuse reflected light, visible breath only if cold enough.
Night/streetlights	Mixed colour temperatures, practical light sources, plausible noise, shadow pools under lights.

F. Prompt Modules

Module	Prompt text
General background replacement	Replace the background with a realistic [location] scene. Match the original camera height, focal length, perspective, lighting, contact shadows, grain, and depth of field. Keep the subject unchanged.
Location accuracy	Use geographically plausible architecture, roads, sidewalks, signage, trees, and weather for [location/date]. Do not add landmarks or storefronts that do not belong there.
Street realism	Add everyday imperfections: uneven pavement, subtle dirt, realistic curb lines, parked vehicles or pedestrians only if scale and blur match. Avoid cinematic fantasy lighting.
Seasonal realism	Make all seasonal cues consistent: clothing, trees, sky, pavement, snow/rain, and light quality must match [season/date].

Quality-Control Procedure

- Does the background match the requested location at an everyday level?
- Do season, weather, clothing, and road conditions agree?
- Are horizon, vanishing points, and camera height coherent?
- Are shadows, contact points, and occlusion believable?
- Is any signage readable or safely blurred?
- Does the subject look photographed there, not pasted on?

Quick Reference Summary

Believable background replacement is camera matching plus context matching. Geography alone is not enough: perspective, shadows, season, weather, architecture, road layout, signage, and grain must all agree.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.
- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.
- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.